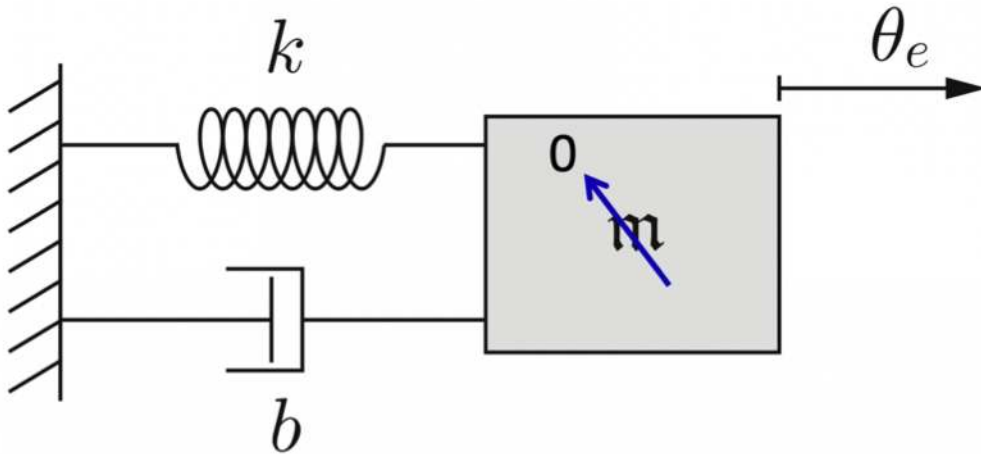


Chapter 2	Configuration Space
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
	11.1 Control System Overview
	11.2 Error Dynamics
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

First-order error dynamics



$$m\ddot{\theta}_e + b\dot{\theta}_e + k\theta_e = 0$$
$$\dot{\theta}_e(t) + \frac{k}{b}\theta_e(t) = 0$$

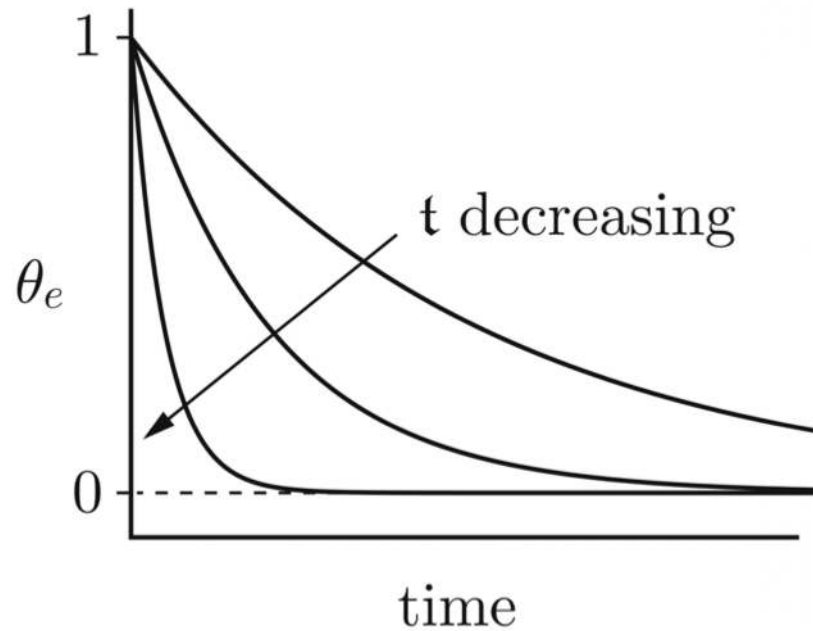
standard first-order form

time constant

$$\tau = b/k$$
$$\dot{\theta}_e(t) + \frac{1}{\tau}\theta_e(t) = 0$$

Important concepts, symbols, and equations (cont.)

First-order error dynamics



$$\theta_e(t) = e^{-t/t} \theta_e(0)$$

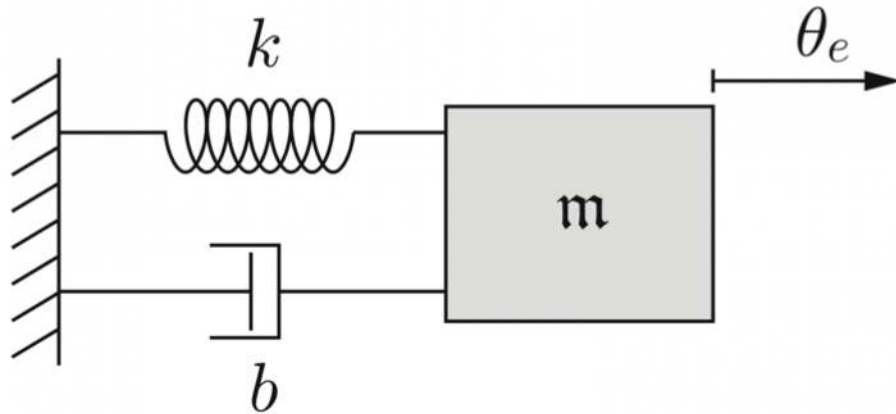
$$\theta_e(0) = 1$$

$$\frac{\theta_e(t)}{\theta_e(0)} = 0.02 = e^{-t/t}$$

$$\ln 0.02 = -t/t \rightarrow t = 3.91t$$

Important concepts, symbols, and equations (cont.)

Second-order error dynamics



$$\ddot{\theta}_e(t) + \frac{b}{m}\dot{\theta}_e(t) + \frac{k}{m}\theta_e(t) = 0$$

natural frequency

damping ratio

$$\omega_n = \sqrt{k/m} \quad \zeta = b/(2\sqrt{km})$$

$$\ddot{\theta}_e(t) + 2\zeta\omega_n\dot{\theta}_e(t) + \omega_n^2\theta_e(t) = 0$$

standard second-order form

Important concepts, symbols, and equations (cont.)

Second-order error dynamics

$$\ddot{\theta}_e(t) + 2\zeta\omega_n\dot{\theta}_e(t) + \omega_n^2\theta_e(t) = 0$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \quad \text{characteristic equation}$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

$\zeta > 1$: Overdamped

$\zeta = 1$: Critically damped

$\zeta < 1$: Underdamped

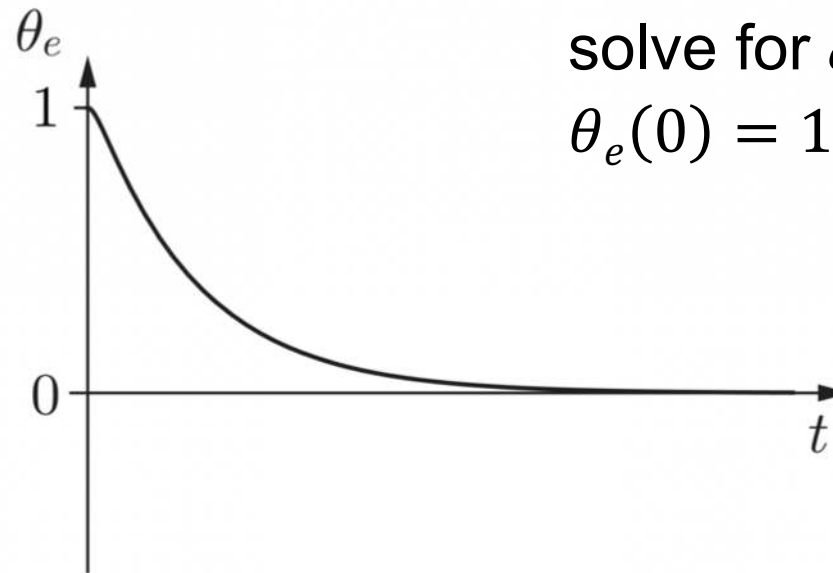
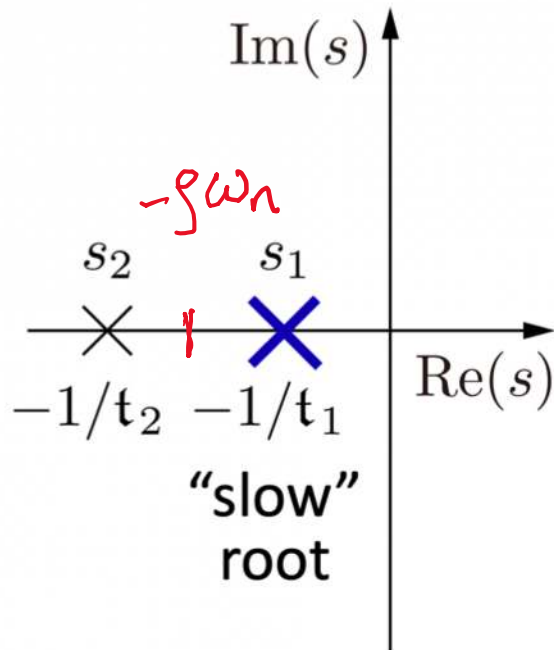
Important concepts, symbols, and equations (cont.)

$\zeta > 1$: Overdamped

$$\theta_e(t) = c_1 e^{s_1 t} + c_2 e^{s_2 t}$$

$$s_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$$

$$s_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$$



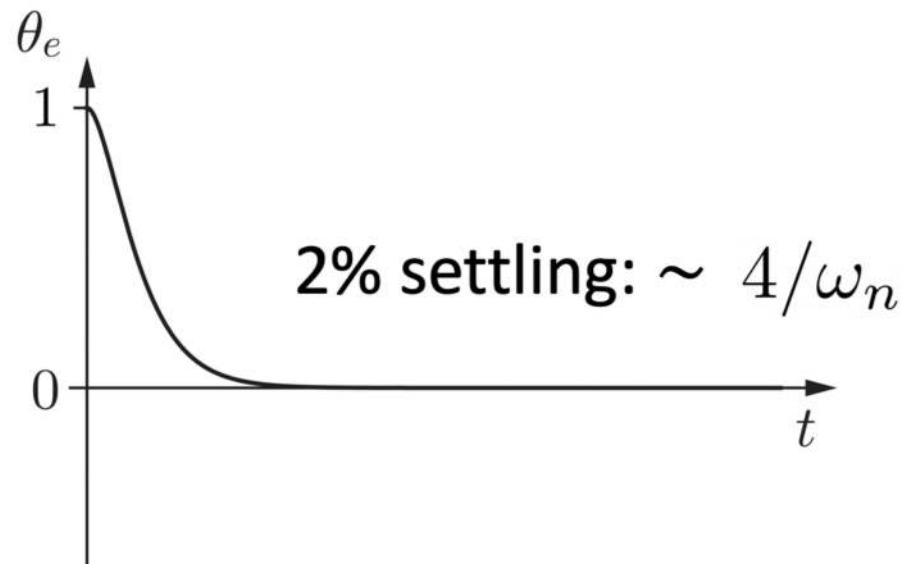
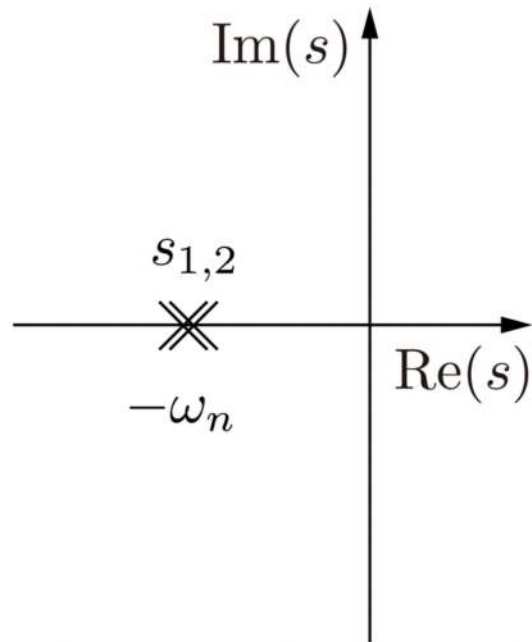
solve for c_1 and c_2 using
 $\theta_e(0) = 1, \dot{\theta}_e(0) = 0$

Important concepts, symbols, and equations (cont.)

$\zeta = 1$: Critically damped

$$\theta_e(t) = (c_1 + c_2 t)e^{-\omega_n t}$$

$$s_{1,2} = -\omega_n$$



Important concepts, symbols, and equations (cont.)

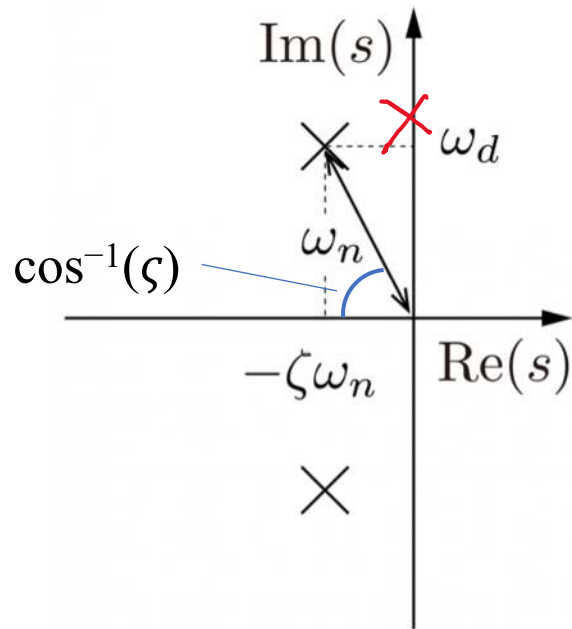
$\zeta < 1$: Underdamped

$$\theta_e(t) = (c_1 \cos \omega_d t + c_2 \sin \omega_d t) e^{-\zeta \omega_n t}$$

damped natural frequency

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$s_{1,2} = -\zeta \omega_n \pm j \omega_d$$

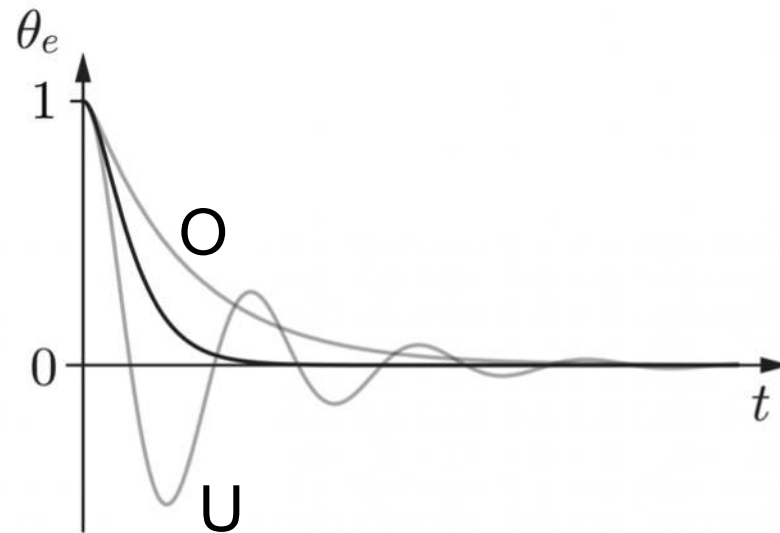
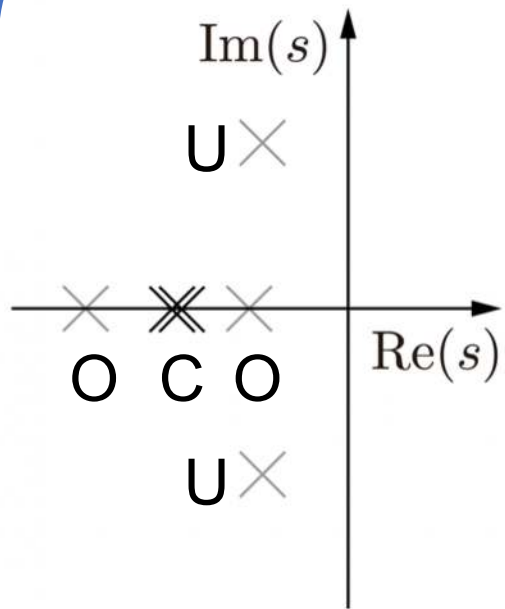


2% settling: $\sim 4 / \zeta \omega_n$

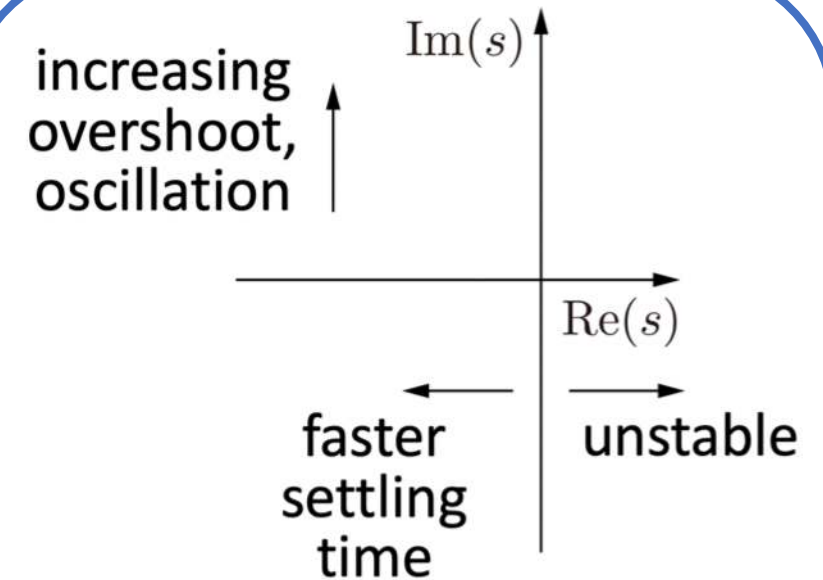
overshoot: $e^{-\pi \zeta / \sqrt{1 - \zeta^2}} \times 100\%$

small $\zeta \rightarrow$ large overshoot

Important concepts, symbols, and equations (cont.)



second-order systems



time response relative
to root locations for
general systems

$$\omega_n = \sqrt{k/m} \quad \zeta = b/(2\sqrt{km})$$

$$\gamma \omega_n = \frac{b}{2\sqrt{km}} \sqrt{\frac{k}{m}} = \frac{b}{2m}$$

2% settling: $\sim 4/\zeta \omega_n \rightarrow \frac{8m}{b}$

$$\ddot{\theta}_e(t) + 2\zeta\omega_n\dot{\theta}_e(t) + \omega_n^2\theta_e(t) = 0$$

overshoot: $e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$

When controlling a robot joint, what do b , k , and m usually correspond to?

m : mass/inertia of link

k : P in PD control

b : D in PD control (+ maybe friction)

How do you change m to decrease settling time? k , b ?

$m \downarrow$ k no effect $b \uparrow$

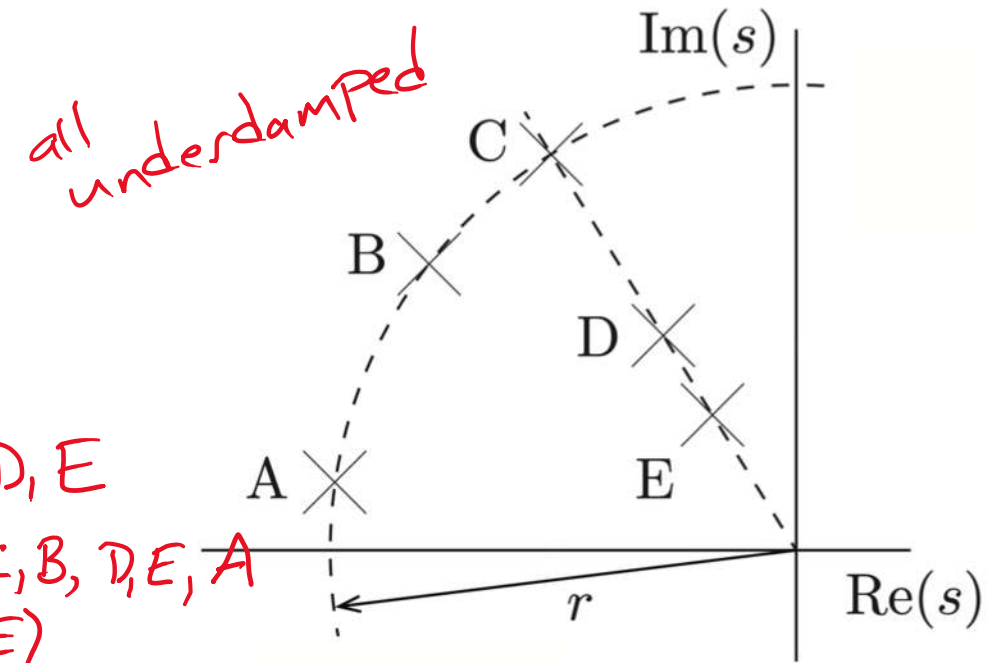
How do you change m to decrease overshoot? k , b ?

increase γ $m \downarrow$ $k \downarrow$ $b \uparrow$

Shown are one of the roots of five different second-order systems, A, B, C, D, and E.

List them in the following orders:

1. Natural frequency, highest to lowest. *(A, B, C), D, E*
2. Damped natural frequency, highest to lowest. *C, B, D, E, A*
3. Damping ratio, highest to lowest. *A, B, (C, D, E)*
4. Overshoot in unit step error response, highest to lowest. *(C, D, E), B, A*
5. Settling time, longest to shortest. *E, D, C, B, A*



Which has the “best” transient response?

*A. Damping ratio is largest, so least overshoot.
Furthest left, so fastest settling time.*